UNDERSTANDING MATANO FAULT, SOUTHEAST ARM OF SULAWESI, INDONESIA

Muhamad Nasir Lukman, Kharis Adyatma Lamaliwa, Salahuddin Husein, Agung Setianto, Saptono Budi Samodra, Sugeng Sapto Surjono

Department of Geological Engineering, Faculty of Engineering, Universitas Gadjah Mada, Yogyakarta, Indonesia

Abstract: Matano Fault was a strike-slip fault with ESE – WNW orientation which located on the northern part of Sulawesi's Southeast Arm. The fault is connected to Palu-Koro Fault on the western part and Sorong Fault on the eastern part. The Digital Elevation Model and the geological map shows that Matano Fault was a left stepping left-lateral fault that formed Matano's Lake pull-apart basin. The Offset of the Tokala Formation shows Matano Fault have at least 60 km displacement. The tectonic stress inversion from earthquake focal mechanism data showsMatano Fault was a strike-slip faultwith N 38° E/ 30° maximum compressive principal stress axes.

Key words: Matano Fault, Sulawesi, Earthquake Focal Mechanism

I. Introduction

Matano Fault was a strike-slip fault with ESE – WNW orientation which cut Southeast Sulawesi through Matano Lake (Surono, 2010). This fault marked by a large valley at the northern part of Southeast Sulawesi that disrupts local drainage pattern (Hamilton, 1979). Matano Fault is connected to Palu-Koro Fault on the western part and Sorong Fault on the eastern part (Figure 1) (Effendi, 1997). This fault was activated/reactivated after the collision event during Early Miocene (Surono, 2010) and still seismically active until today (Hamilton, 1974 from Hamilton, 1979; Surono, 2010). This research aims to get more thorough understanding about the nature of the fault, especially about the sense and the geometry of the fault, along with the maximum compressive principal stress that acted on the fault.

This fault located on the Sulawesi Ophiolite Belt that uplifted during the collision event at Late Oligocene (Surono, 2010). Several other major faults that located on the

Southeast Arm of Sulawesi such as Lawanopo Fault and Kolaka Fault (Figure 1)also have the same orientation and movement sense with Matano Fault.

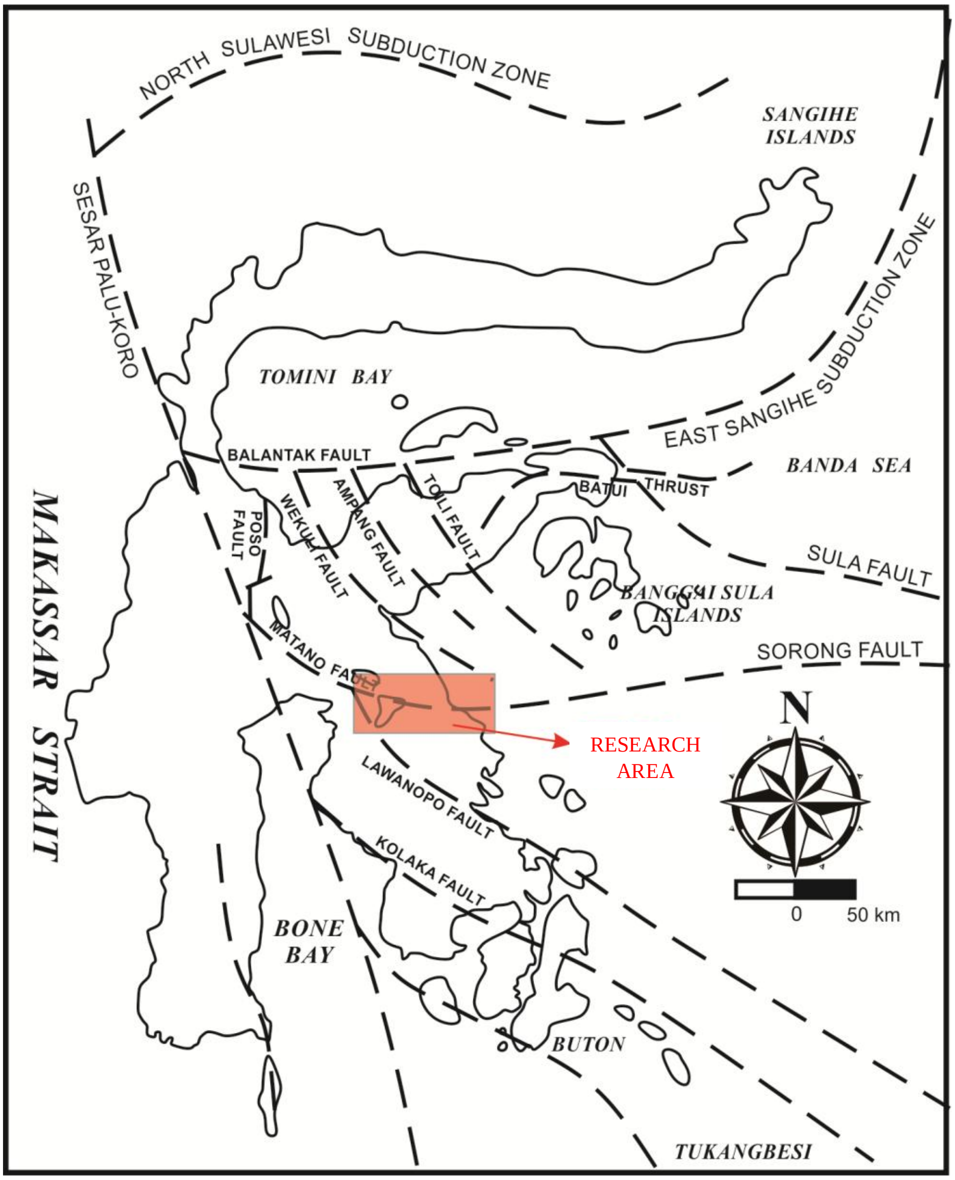


Figure 1. Map showing major geological structure around Sulawesi and the research area (after Effendi, 1997)

II. Methods

This research applied remote sensing method on Digital Elevation Model derived from IFSAR imagery and geological map to study the geometry of the fault and the surrounding area, followed by the tectonic stress inversion analysis of the earthquake focal mechanism data to understand the movement sense of the fault and to obtain the maximum compressive principal stress acted on the fault.

III. Results and Discussion

Matano Fault can be recognized from the Digital Elevation Model as a big valley that cut the hilly terrain of the northern part of the area (Figure 2). The evidence of Matano's Fault left-lateral movement can be observed on Figure 2 as the offset of major geological units, such as the offset of Tokala Formation located on the eastern coast of Sulawesi's Southeast Arm. The apparent offset of Tokala Formation suggest that Matano Fault's displacement is roughly 60 km long.

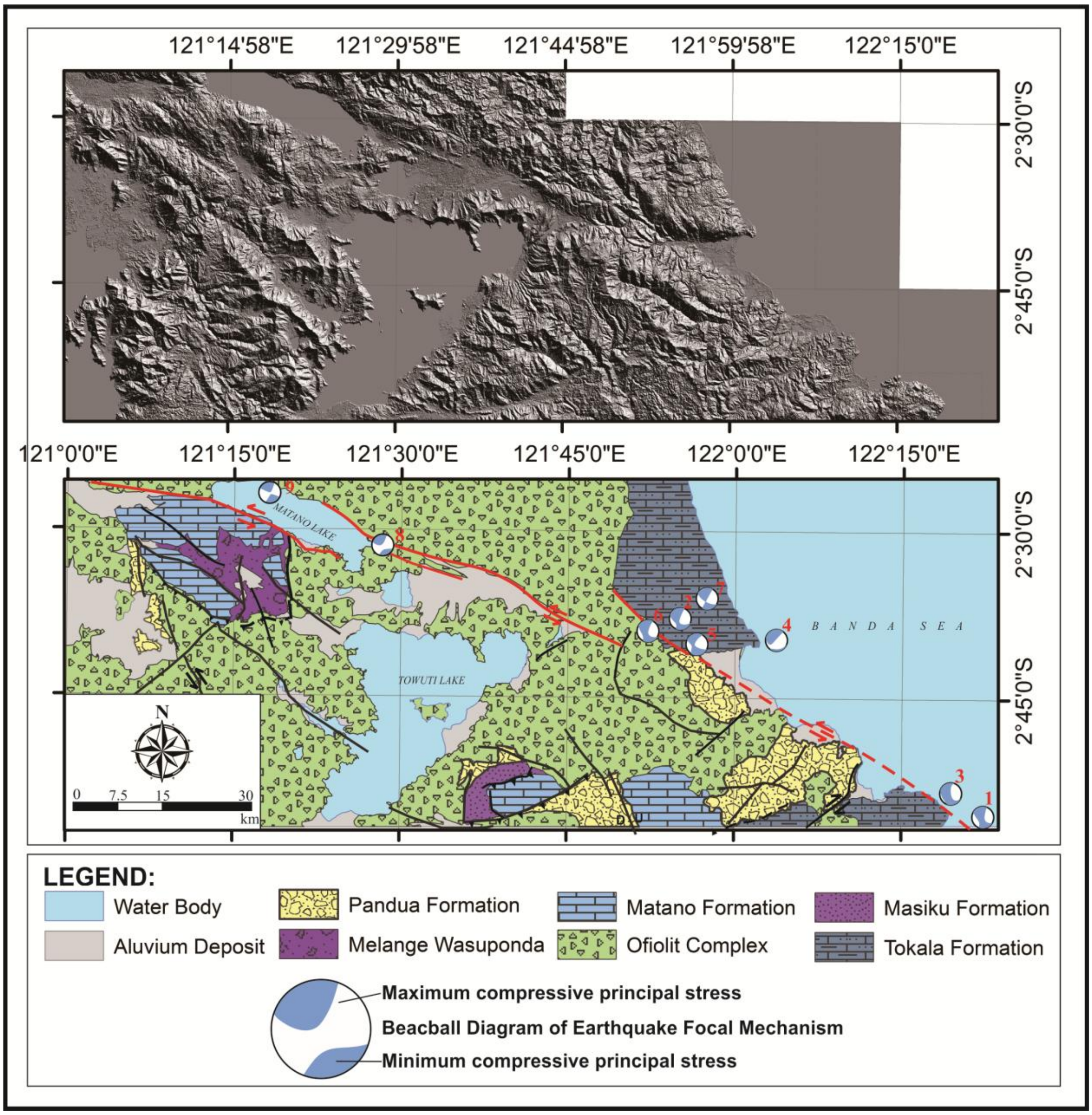
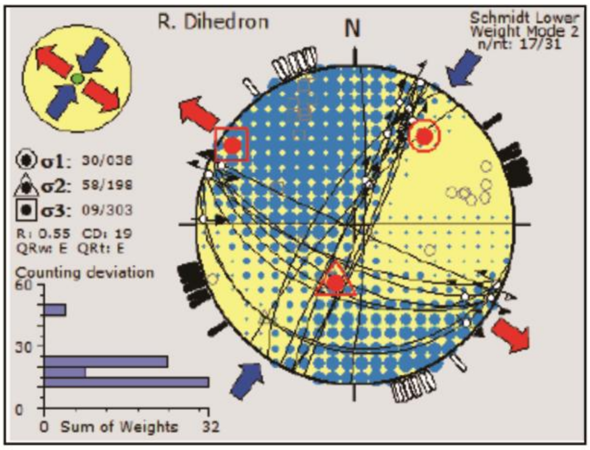


Figure 2. Digital Elevation Model and Geological Map of the area surrounding Matano Fault

Matano Fault is not a continuous fault. As showed onFigure 2, Matano Fault seems to be a segmented fault with a left-stepping characteristic. This left-stepping characteristic of left-lateral strike-slip fault generated a transtension zone, which in this case formed a pull-apart basin such as Matano Lake.

The Earthquake data gathered from USGS Earthquake Archives suggested that Matano Fault is still a seismically active fault. The earthquake data from 1970 shows 42 earthquake above 4.5 RS located around the fault. From those data, there are 9 earthquake that contains a focal mechanism data that shown at the map on Figure 2 and at the table on Figure 3.The tectonic stress inversion from all of the focal mechanism data (Figure 3) analyzed with Win_Tensor version 5.0.7 shows the strike-slip nature of the fault with a maximum compressive stress axes at N 38° E/ 30° (NE – SW orientation of maximum compressive stress axes).



No.	Time	Depth (km)	Mag (RS)
1	December 2014	10	5.5
2	May 2012	10	5.1
3	November 2001	33	5.3
4	November 1980	46	5.6
5	May 2000	33	5.2
6	April 2012	13	5.8
7	April 2012	10	5.3
8	February 2011	16.2	6.1
9	June 1995	63.7	5.3

Figure 3. Result of tectonic stress inversion from 9 earthquake focal mechanism data (left); Table that shows the additional data of the 9 earthquake focal mechanism

III. Conculsion

Matano Fault was a left stepping left-lateral fault that have at least 60 km displacement. The tectonic stress inversion from earthquake focal mechanism data shows Matano Fault was a strike-slip fault with N 38° E/ 30° maximum compressive principal stress axes.

References

Delvaux, D., B. Sperner, 2003. *New Aspects of Tectonic Stress Inversion With Reference to the TENSOR program.* New Insight into Structural Interpretation and Modelling, Special Publications, 212, h. 75-100. Geological Society, London

Effendi, A.C.., S.S. Bawono, 1997. *Peta Geologi Lembar Manado, Sulawesi Utara, skala 1 : 250.000.* Pusat Penelitian dan Pengembangan Geologi, Bandung

Hamilton, W.B., 1979. *Tectonics of the Indonesian Region.* United States Geological Survey Professional Paper 1078, Washington, D.C

Simandjuntak, T.O., E. Rusmana, J.B. Supandjono, A. Koswara, 1993. *Peta Geologi Lembar Bungku, Sulawesi, skala 1 : 250.000.* Pusat Penelitian dan Pengembangan Geologi, Bandung

Stein, S., Michael W., 2003. *An Introduction to Seismology, Earthquakes, and Earth Structure* Blackwell Publishing Ltd, United Kingdom

Surono, 2010. *Geologi Lengan Tenggara Sulawesi*. Badan Geologi Kementerian Energi dan Sumberdaya Mineral, Bandung